

Updated Research Proposal

Title

Evaluating South Africa's Readiness for Integrating Software as a Medical Device (SaMD) and AI as a Medical Device (AIaMD): Framework-Based Gap Analysis, Stakeholder Assessment, and Operational Recommendations

Background and Motivation

Advanced digital health technologies, specifically Software as a Medical Device (SaMD) and AI-driven medical devices (AIaMD), offer transformative potential for healthcare improvement in South Africa. Despite this potential, regulatory challenges persist due to the rapid evolution of software-based medical solutions. Current regulations under the South African Health Products Regulatory Authority (SAHPRA) predominantly focus on hardware-centric devices, neglecting unique aspects such as continuous software updates, cybersecurity threats, and algorithmic bias.

Problem Statement

There is an evident regulatory gap regarding the oversight of SaMD and AIaMD within South Africa's existing medical device regulatory framework. Critical issues such as algorithmic transparency, data privacy, bias, and system robustness remain inadequately addressed, creating barriers for the safe, effective, and ethical deployment of these advanced technologies.

Research Aim

To systematically assess South Africa's regulatory readiness for SaMD and AIaMD adoption using a structured gap analysis framework, complemented by comprehensive stakeholder assessments, and to develop operational recommendations to bridge identified gaps.

Research Objectives

1. **Framework-Based Gap Analysis:** Evaluate existing South African regulations using established global frameworks (e.g., FDA, EU MDR, WHO guidelines) to identify specific regulatory deficiencies.
2. **Stakeholder Consultation:** Engage with key stakeholders through structured questionnaires and interviews to capture perceptions, practical experiences, and expectations regarding regulatory oversight.
3. **Comparative Analysis:** Compare framework-based analysis results with stakeholder input to clearly highlight regulatory gaps.

4. **Recommendations Development:** Formulate actionable recommendations and develop an operational compliance tool aligned with international best practices and stakeholder needs.

Research Questions

- What specific gaps exist within SAHPRA's regulatory framework concerning accountability, fairness, human oversight, transparency, and technical robustness?
- How do stakeholder perceptions align or diverge from academic and framework-based regulatory gap analyses?
- What practical, implementable strategies can effectively address the identified regulatory gaps?

Methodology

Framework-Based Gap Analysis

- Utilize frameworks such as FDA Good Machine Learning Practices, EU MDR, and WHO global guidelines to evaluate SAHPRA's regulatory framework.
- Assess regulatory dimensions including accountability, fairness, human agency, privacy, technical robustness, transparency, and societal impact.

Stakeholder Consultation

- Develop structured questionnaires and semi-structured interview protocols aligned with the selected regulatory evaluation frameworks.
- Conduct interviews and focus groups with SAHPRA officials, ethics committees, healthcare professionals, technology developers, and industry experts.

Comparative Analysis

- Identify areas of convergence and divergence between stakeholder experiences and framework-based assessments.
- Clearly outline prioritized gaps based on stakeholder and framework analyses.

Recommendations and Compliance Tool

- Generate actionable, prioritized recommendations addressing identified gaps.
- Develop an operational compliance tool reflecting stakeholder realities, academic insights, and international regulatory best practices.

Expected Outcomes

- Comprehensive regulatory gap analysis highlighting critical areas needing attention.

- Detailed stakeholder insights, reflecting practical challenges and perceived regulatory needs.
- Clear, actionable recommendations and an operational compliance tool facilitating improved regulatory clarity, safety, and innovation for SaMD and AlaMD in South Africa.

Significance of the Study

This study directly addresses critical regulatory gaps, promotes safer integration of advanced medical technologies, supports innovation among healthcare technology startups, and positions South Africa competitively within the global digital health landscape. It enhances transparency, ethical governance, and practical applicability of regulatory frameworks, fostering trust and safety in AI-driven medical solutions.